

Far from the frontier: the China shock and firm innovation in Latin America

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Abstract

The rise of China in world trade exposed manufacturing firms to two shocks at once: sharper competition in the markets they sell into, and cheaper and more sophisticated inputs in the markets they buy from. This paper separates the two and estimates their effects on firm innovation in Argentina, Brazil and Mexico. Combining patent data for 2,576 innovating firms with bilateral trade flows decomposed through inter-country input-output and national supply-use tables, I measure each country-industry's exposure to China on both sides of the transaction at the 4-digit level, and instrument each with China's rise in donor economies that exclude the treated country. Import competition reduced patenting; input access raised it, by substantially more. The response concentrates among technologically intensive firms and runs through technologically intensive intermediates, a pattern consistent with firms operating far from the frontier learning from what imported inputs embody. These results also suggest that single import-penetration measures, by bundling a potentially discouraging horizontal shock with an enabling vertical one, account for part of the literature's disagreement over the effects of Chinese imports on firm performance.

Keywords: Latin America, China shock, innovation, input access, import competition

JEL classification: F14, F60, L60, O31, O34

1 Introduction

Over the past quarter century, China’s integration into the global trading system has been one of the most consequential developments in the world economy. This episode, reflecting the emergence of China as a dominant player in global manufacturing after its accession to the World Trade Organization in late 2001, has been often referred to as the “China shock” in the economics literature (Autor et al., 2016). As such, it was perceived in all channels of trade exposure: increasing import competition, access to imported inputs, and market access to exports. In just six years, from 2002 to 2008, China’s weight as an exporter nearly doubled, representing from 4.1% to 7.6% of world exports. As an importer, its weight rose from 3.7% to 5.9% in the same period.¹ For firms around the world, the surge in Chinese exports meant sharper competition from Chinese goods in their own product market, but also cheaper and higher-quality inputs from Chinese suppliers upstream — two forms of exposure that had distinct consequences for producers.

This distinction matters because it can reconcile a literature that has not settled on a sign. Studies of the innovation response to Chinese imports find positive effects in some settings (Ahn et al., 2018; Bloom et al., 2016) and negative effects in others (Aghion et al., 2024; Autor et al., 2020; Cusolito et al., 2023). Part of this disagreement may be definitional. A single net import-penetration measure conflates a horizontal competition shock, which Schumpeterian logic predicts should discourage innovation among firms far from the frontier, with a vertical input-access shock that lowers costs and transmits embodied technology. Where the two are bundled into one regressor, the estimated effect depends on which channel happens to dominate in the data. Separating them is necessary to interpret the net effect, and doing so requires input–output data that map each industry’s exposure on both sides.

This paper provides such a decomposition for Latin America. I study how the China shock affected firm innovation, measured by patenting, in Argentina, Brazil, and Mexico, splitting each country-industry’s exposure to China into an import competition channel and an input access channel built from layered input–output data at the 4-digit industry level.² Evidence of this kind is scarce for emerging economies: analyses of the China shock on firms have focused on developed countries (Aghion et al., 2024; Autor et al., 2020; Bloom et al., 2016), while most studies of trade and firm performance in emerging markets examine earlier liberalization episodes and focus on

¹Figures calculated with World Bank data.

²China’s demand for Latin American goods fell overwhelmingly on commodities rather than on manufactures, which is why this study does not include evidence on the increased market access channel.

productivity rather than innovation ([Amiti and Konings, 2007](#); [Bas and Ledezma, 2010](#); [Schor, 2004](#)). The three countries I study account for over 85% of all patenting activity in Latin America between 1996 and 2008.

The region is not merely a further setting in which to run the decomposition; it is one in which the two channels are predicted to separate more sharply than elsewhere. Latin American firms operate well behind the global technological frontier, a gap that is long-standing and did not narrow in the decades preceding the shock ([Cole et al., 2005](#); [Daude and Fernández-Arias, 2010](#)). Both mechanisms turn on that distance, and in opposite directions. Where post-innovation rents are remote, the escape-competition response to import competition is weak and additional competitive pressure is more likely to depress innovative investment than to spur it ([Aghion et al., 2005](#)). Where the technology embodied in imported intermediates is far from what domestic firms already command, the scope for learning and upgrading through inputs is correspondingly larger. The same feature of the setting that sharpens the negative channel sharpens the positive one, which is what makes the region informative about their relative weight.

The closest paper to this study is [Aghion et al. \(2024\)](#), which disentangles import competition from input access using French firm-level accounting and customs data and finds opposing effects: competition detrimental to innovation, and the input-supply shock positive, though not always significant. I extend this decomposition to a developing region setting that lacks firm-level customs microdata, across three countries, for firms far from the frontier. This also separates the paper from work that isolates a single margin — competition, in [Cusolito et al. \(2023\)](#) for Chile — or that distinguishes the channels through tariff variation rather than within a single trade shock ([Bas and Paunov, 2021](#)). Methodologically, the input–output strategy follows [Acemoglu et al. \(2016\)](#), who use national input–output tables to trace the China shock through inter-industry linkages and find that these network effects substantially amplify the direct employment losses.

Building the two exposures requires trade data that identify not only the exporting industry of a flow but also the industry that uses it, which standard bilateral sources do not report. I construct this by combining product-level trade data with inter-country input–output tables and national supply-use tables, yielding, for each country-industry, both the inflow of competing goods and the inflow of intermediates from every supplying industry. The two measures are built from the same underlying flows viewed from opposite sides of the transaction, which is precisely why they must be estimated jointly. A Chinese export surge raises competition for the firms producing those

goods and improves input access for the firms buying them, and it does so simultaneously. Both exposures are endogenous to domestic conditions, since an industry drawing in imports because its own demand is expanding looks much the same in the data, so I instrument each with China’s rise as a supplier to donor economies that exclude the treated country, following [Autor et al. \(2020\)](#). The sample comprises 2,576 innovating firms responsible for 8,488 patent families over 1996–2008.

The two channels enter with opposite signs, and the vertical one dominates. A one percentage point increase in China’s share of a country-industry’s imports of competing goods reduces average annual patenting, while an equivalent increase in China’s share of its imported intermediates raises it by roughly five times as much; evaluated at one standard deviation of each exposure, input access is the larger force by a factor of about two and a half. Estimating the channels jointly matters for the competition estimate as well as for the input estimate: the competition coefficient grows in absolute value once input access is held constant, which is the confounding a bundled penetration measure induces, shown within a single dataset.

The response is concentrated rather than diffuse. Splitting firms by the technological intensity of their own sector, both effects concentrate among high-technology firms, and the difference between the groups is itself significant. Splitting input exposure by the technological intensity of the supplying industry, the effect is carried entirely by high-technology intermediates, while low-technology inputs, the larger inflow to the region by volume, show no measurable response. The pattern is consistent with a learning mechanism, though it does not decisively separate learning from a cost channel operating through the same inputs. Citations per patent respond positively to input access, suggesting that the additional patenting was not simply defensive protection of existing intellectual property, and the results survive replacing the growth-rate outcome with patent counts, weighting firms by their patent stock, and restricting the sample to firms patenting in both periods.

The remainder of the paper is organized as follows. Section [2](#) reviews the relevant theory and empirical literature on trade and innovation. Section [3](#) describes how the China shock unfolded in Latin America and why the region’s distance from the technological frontier shapes what to expect. Section [4](#) sets out data sources and construction, with descriptive statistics. Section [5](#) outlines the empirical strategy, and Section [6](#) presents results. Section [7](#) concludes.

2 Theoretical motivation

Innovation can take multiple forms for firms. While its main input is generally agreed to be investment in R&D, its outcome, if successful, can be the adoption of new production processes, upgraded product quality, or an entirely new set of products. The intellectual property of these outcomes is often protected by patents, which is why measuring innovation directly through patenting has been a common approach in the literature ([Akcigit and Melitz, 2022](#)). Innovation sits within the broader literature on firm performance, most of which focuses on productivity ([Shu and Steinwender, 2019](#)); innovation ultimately affects productivity, but the translation is slow and productivity gains can arise from processes that do not involve innovation, which motivates measuring it directly.

Investment in R&D, which may or may not yield a successful innovation, is determined by firms as a function of the expected future profits from that innovation. Because it carries high fixed costs, it benefits from economies of scale: [Aghion et al. \(2024\)](#) document multiple innovation premia, showing that French firms that both export and innovate are substantially larger and more productive than non-innovating exporters. These expected profits depend in turn on product market competition, which trade liberalization can reshape ([Melitz and Redding, 2021](#)).

A firm is simultaneously a seller of its output and a buyer of its inputs, and trade exposure can act on both functions. On the selling side, an inflow of imports of goods similar to those the firm produces intensifies *import competition*, a horizontal shock to its product market. On the buying side, an inflow of imported inputs the firm uses in production expands its *input access*, a vertical shock to its supply chain ([Shu and Steinwender, 2019](#)). These two channels — one acting on competitive pressure, the other on production technology — operate through imports into the domestic economy and bear most directly on the innovation incentives of manufacturing firms. They are the focus of this paper.³ The next two subsections discuss each in turn.

2.1 Import competition

The relationship between competition and innovation has been part of a long-standing debate in economics. Competition can be detrimental to innovation by restricting firm size and weaken-

³A third channel, export market access, expands a firm’s selling market abroad and can likewise affect innovation, with mixed evidence ([Ahn et al., 2018](#); [Bustos, 2011](#); [Coelli et al., 2022](#); [Ghizzi, 2021](#)). In the Latin American setting I study, however, this channel operates mainly through commodity exporters rather than the manufacturing firms whose patenting I analyze. For this reason, this study does not aim to provide evidence on the export market access channel.

ing economies of scale (Schumpeter, 1943); this Schumpeterian force is countered by the escape-competition effect (Arrow, 1962), in which greater competition erodes incumbents' pre-innovation rents and thereby raises the incremental return to innovating. Aghion et al. (2005) reconcile the two with a nonlinear relationship that depends on the productivity dispersion of an industry: where firms are neck-and-neck, the escape-competition effect dominates, whereas in leader-laggard industries — those whose firms sit far from the productivity frontier — an additional competition shock triggers the Schumpeterian effect and depresses innovation. This distance-to-frontier logic is central to our setting. Cusolito et al. (2023) test it using a matched production-innovation panel for Chile, finding that the China shock depressed most measures of firm innovation, with gains confined to the most productive decile of plants and the negative response sharpest where pre-shock rents and proximity to the frontier were lowest. Because Latin American firms typically operate well behind the global frontier, the framework predicts that import competition is more likely to discourage than to spur their innovation.

Empirically, much as the theory implies, the cross-country evidence is mixed. One group of studies finds a positive relationship between import competition and innovation — in Europe (Bloom et al., 2016), Korea (Ahn et al., 2018), and Japan (Yamashita and Yamauchi, 2020) — consistent with the escape-competition effect. Others find negative effects, in the United States (Autor et al., 2020), Canada (on process rather than product innovation) (Yang et al., 2021), France (Aghion et al., 2024), and Brazil (Ghizzi, 2021; Moreira et al., 2023). In emerging economies the response often operates through firm and product selection rather than within-firm intensity: facing Chinese competition, Mexican plants shed marginal products and the smallest exit, while larger plants expand (Iacovone et al., 2013), and Indian firms drop peripheral products under domestic competitive pressure, an effect concentrated among smaller firms and producers of intermediate goods (Chakraborty and Henry, 2019). Across these settings, a robust regularity is that larger, more productive firms fare relatively better — or at least less badly — than laggards under an import competition shock.

2.2 Input access

Unlike import competition, which is a horizontal shock from similar goods, access to imported inputs is a vertical shock from goods used in the firm's own production, and theory predicts it should benefit performance and innovation. The gains run through two main distinct channels.

The first is a cost channel: cheaper intermediates raise profits and the resources available for R&D (Shu and Steinwender, 2019), an effect especially salient for inputs from emerging suppliers such as China in the early 2000s. The second is a learning channel: access to higher-quality or more varied foreign inputs transmits embodied technology and knowledge spillovers (Grossman and Helpman, 1991), raising the firm’s own technological capability. The distinction matters for my mechanism: a learning channel implies that the firms with the most to learn — those furthest from the frontier, and in technologically intensive sectors — should gain the most. Evidence from developing economies points to the learning margin: input-tariff liberalization leads firms to source higher-quality foreign inputs and upgrade the quality of their own output (Bas and Paunov, 2021; Bas and Strauss-Kahn, 2015), a pattern Verhoogen (2023) synthesizes as an input-side driver of firm-level upgrading.

Empirical work on input access has expanded as input–output and firm-level data have made it possible to separate inputs from overall imports. The canonical finding is that access to new imported input varieties lets firms introduce new domestic products (Goldberg et al., 2010). The productivity evidence is consistent and positive across emerging economies (Amiti and Konings, 2007; Schor, 2004; Topalova and Khandelwal, 2011), and the evidence on innovation, though often weaker, points the same way, whether measured by patenting (Aghion et al., 2024; Bloom et al., 2016), product innovation (Bas and Paunov, 2021; Colantone and Crinò, 2014), or technology adoption (Bas and Berthou, 2016). Notably, the studies that separate the two channels tend to recover opposing signs — competition contracting product scope while imported intermediates expand it (Chakraborty and Henry, 2019), or output- and input-tariff movements pulling firm decisions in opposite directions (Bas and Paunov, 2021) — which motivates measuring them jointly rather than through a single net import measure.

2.3 Contribution

This paper relates most closely to three strands of literature. A first studies the innovation response to Chinese competition in developing countries, including Latin America (Cusolito et al., 2023; Iacovone et al., 2013; Moreira et al., 2023); a second studies the firm-level gains from imported inputs (Bas and Strauss-Kahn, 2015; Goldberg et al., 2010); and a third, closest in spirit, separates the competition and input-access margins of trade reform (Bas and Paunov, 2021; Chakraborty and Henry, 2019). I depart from each in three ways. First, where prior work isolates one margin

— competition in [Cusolito et al. \(2023\)](#), input access in [Goldberg et al. \(2010\)](#) — or separates the two using tariff instruments ([Bas and Paunov, 2021](#)) or by contrasting total with intermediate imports ([Chakraborty and Henry, 2019](#)), I decompose a bilateral trade expansion into horizontal (import-competition) and vertical (input-access) exposure within one input–output framework, so the two channels are measured on a common footing and their magnitudes are directly comparable. Second, I construct this decomposition at the 4-digit industry level across three Latin American economies, capturing the majority of patenting activity in a continent with relatively fewer innovating firms. Third, by separating the channels I offer a reconciliation of the conflicting estimates in the competition–innovation literature: a single import-penetration measure conflates a discouraging horizontal shock with an enabling vertical one, and which dominates depends on a country’s position in the China trade relationship.

Measuring innovation through patenting faces certain limitations. It is an imperfect proxy, since patenting in developing economies is low and concentrated among larger firms, and richer firm-level innovation surveys can capture margins it misses ([Cusolito et al., 2023](#); [Verhoogen, 2023](#)). I adopt it deliberately: harmonized patent data are comparable across Argentina, Brazil, and Mexico, whereas the innovation-survey measures used in single-country studies are not available on a consistent cross-country basis. A more fundamental concern is whether patents capture genuine innovation rather than firms’ propensity to seek legal protection. Two versions of this concern apply to the two channels differently. The classical lawyer effect—that competition raises patenting without raising innovation, as firms defensively protect existing intellectual property—is coherent for the import-competition channel, where the arrival of rivals gives firms a motive to fence off their position. It is weaker for input access: cheaper or better inputs create no new appropriability threat, and it is unclear what a firm sourcing improved intermediates would be defensively protecting. [Section 6.3](#) provides further evidence against the lawyer effect behind both channels, in particular showing that the input-access response raises patent quality, not only patent counts—consistent with [Coelli et al. \(2022\)](#), who document that quality did not fall as patenting rose during the 1990s liberalization, contrary to what a wave of low-value defensive filings would imply.

3 An overview of the China shock in Latin America

Understanding how the China shock unfolded in Latin America requires situating it against the region’s economic context in the late 1990s. The region had undergone substantial trade liberalisation through the decade, sharply reducing average tariffs.⁴ This reversed the import substitution policies pursued since at least the 1940s, and a process of deindustrialisation was already under way by the time the China shock arrived — the inverse of the trajectory of emerging East Asian economies (Duran et al., 2008). The shock therefore landed on economies that were already open and already shedding industrial activity, rather than triggering that opening itself.

The period was also marked by currency crises across the region, including in Mexico and Brazil and above all in Argentina, whose convertibility crisis brought severe political and social turmoil alongside a contraction of close to a fifth of real GDP between the 1998 peak and the 2002 trough, followed by a swift recovery (Salvia, 2015). Regional innovation was trending downward over these years: Argentina saw the steepest proportional decline, and Brazil too experienced a marked drop in patenting between 2000 and 2002.

Firms far from the technological frontier. Firms in the region also operated well behind the global technological frontier. The productivity gap is long-standing and did not narrow over the decades preceding the shock: Cole et al. (2005) document that Latin American total factor productivity relative to the United States stood at roughly half and diverged rather than converged through the postwar period, and Daude and Fernández-Arias (2010) attribute the bulk of the region’s income gap to productivity rather than to factor accumulation. Pagés (2010) reaches the same conclusion for the region as a whole.

What sustained the gap is not the unavailability of frontier technology. Comin and Mestieri (2018) show that adoption lags across countries narrowed substantially over the twentieth century while the intensity with which adopted technologies are used diverged: technology arrived, and firms used less of it. Cirera and Maloney (2017) formulate the resulting tension as an innovation paradox — the theoretical returns to adoption should be highest precisely where firms are furthest behind, yet firms in developing countries invest least in innovation, because the complementary capabilities that make adoption productive, from managerial quality to technical skills to organisational capital,

⁴The average unweighted applied tariff in Latin America fell from 40% in 1985 to around 13% in 1995 (Duran et al., 2008).

are the binding constraint.

The firms in this paper are not expected to be the average firm in their economies because of innovator premia. But the relevant comparison for both channels is with the global frontier in a firm’s own industry rather than with its domestic peers, and on that comparison even the region’s most capable manufacturers were distant in 2002. This matters both for import competition and input access in Latin America. As established, where post-innovation rents are remote, the escape-competition response to import competition is weak, and additional competitive pressure is more likely to depress innovative investment than to spur it (Aghion et al., 2005). Where the technology embodied in imported intermediates is far from what domestic firms already command, the scope for learning and upgrading through inputs is also correspondingly larger. Latin America is therefore not a geographic replication of the settings in which these channels have mostly been studied, but one in which they are predicted to separate more sharply.

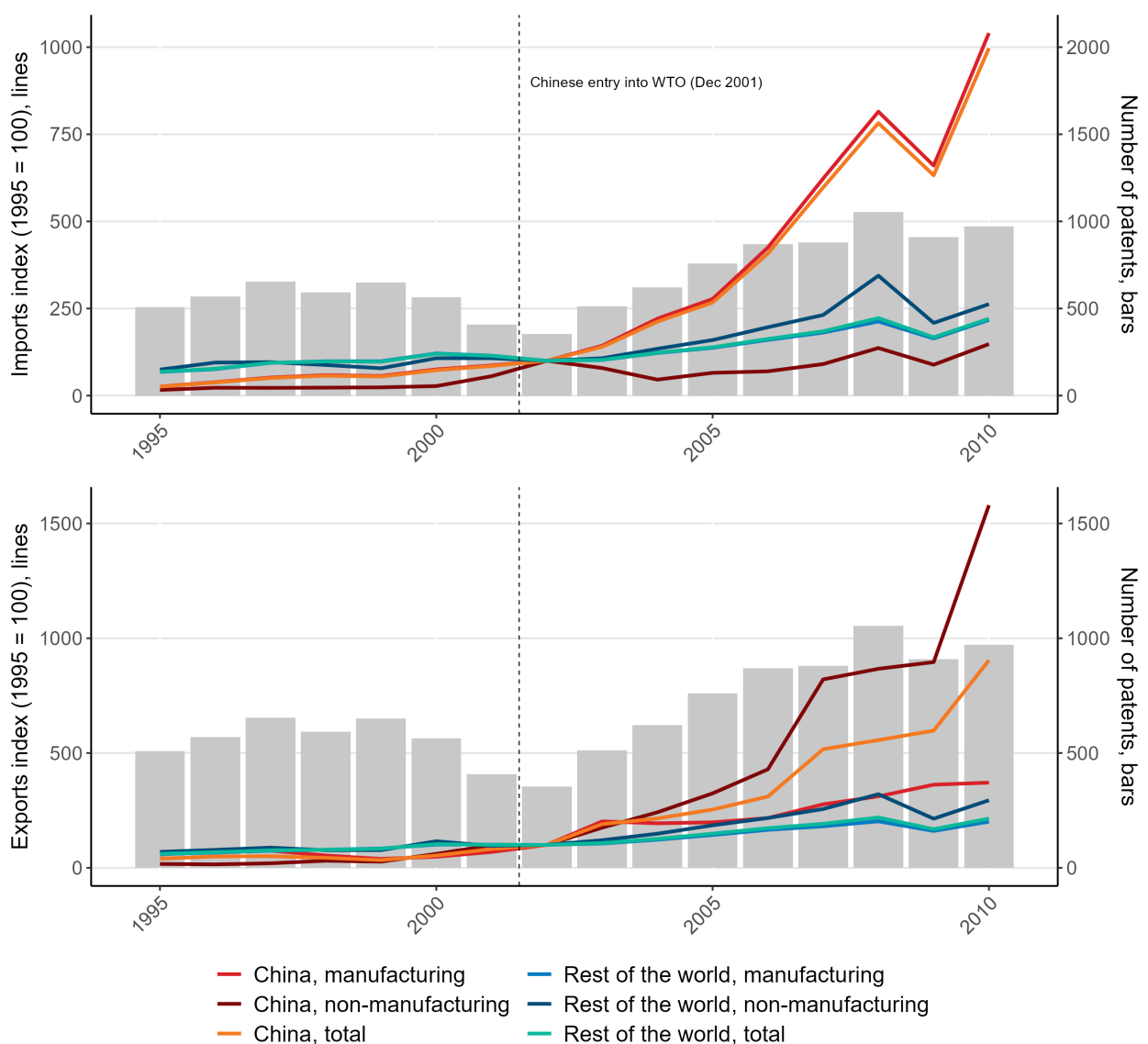
The shape of the shock. Figure 1 presents the evolution of patenting in Argentina, Brazil and Mexico between 1995 and 2010 alongside indexes of import and export growth, split by partner (China and the rest of the world) and by sector (manufacturing and non-manufacturing). A temporary downturn appears in 2009 with the financial crisis, when trade with China recovered faster than trade with the rest of the world; to avoid capturing this episode, I define the period of the China shock as 2002 to 2008.

Two features of the figure organise the rest of the paper. The first is the asymmetry between China and the rest of the world as a source of Latin American imports. Manufactured imports from China outgrew those from the rest of the world by nearly a factor of seven over 2002–2008 (715% against 113%), while for non-manufactured imports China was outpaced. The inflow during the shock was overwhelmingly of manufactured goods, and China entered the region above all as a supplier of manufactures — generating intense exposure on both import-side channels at once: competition for the firms producing similar goods, and input access for those buying them.

The second feature is the contrasting picture on the export side, and it is the reason this paper sets the export channel aside. China’s demand shock to Latin America translated mostly into demand for non-manufactured goods — a commodities-for-manufactures pattern that distinguishes the region from the developed economies studied in much of the literature, where China’s rise also increased demand for manufactured exports. Latin American manufactured exports outpaced rest-

of-world growth by a factor of two, but non-manufactured exports did so by nearly a factor of four; China's weight rose sharply in the region's manufactured imports and barely at all in its manufactured exports. This demand boom had real welfare consequences, including wage growth for agricultural workers (Costa et al., 2016), but it bears little on the firms whose patenting I study, which are essentially all in manufacturing. Export market access is therefore not part of the analysis that follows: the trade data of Section 4 are constructed from import flows, and the empirical strategy of Section 5 estimates the two import-side channels.

Figure 1: The rise of China and innovation in Latin America, 1995-2010



Source: Author's calculations based on BACI (CEPII), combined with BTDIxE (OECD).

Note: Manufacturing corresponds to ISIC Rev.4 industries 10 to 33. Non-manufacturing corresponds to the residual trade, mainly from the agriculture and extractive sectors.

4 Data

With the goal of analyzing the impact of the China shock on Latin America across distinct forms of trade exposure, I combine firm-level patenting data with trade data at the 4-digit industry level. The trade data corresponds to a novel dataset of bilateral trade data that also accounts for the trade of intermediates between China, Latin America and the rest of the world. To achieve this, I combine product-level trade data with a well-established world input-output table dataset, benefitting from extended industry granularity based on national supply-use tables. The final data runs from 1996 through 2008, accounting for the pre-shock period of 1996-2001, and the shock period from 2002 to 2008.

As it aims to draw insights from patenting activity throughout Latin America, the database is cross-country and covers the three main patenting countries in the period: Brazil, Argentina and Mexico. According to PATSTAT data, these countries together represent approximately 85% of all granted patents and innovating firms in Latin America in the period. This approach enables further analysis in a region where patenting is much sparser than in developed economies. Doing so maximizes the number of observed markets (in this study, country-industry pairs) distinctly affected by the China shock.

4.1 Patent data

I draw patent data from the European Patent Office’s Worldwide Patent Statistical Database (PATSTAT), Autumn 2022 edition, which provides bibliographic data, family links, and technology classifications for over 100 million patent documents worldwide. From this source I extract all applications whose applicant address is recorded in Argentina, Brazil, or Mexico and whose applicant is classified as a company in PATSTAT’s sector field, retaining for each application the applicant firm’s name and country, filing date and authority, technology area (IPC code), grant status, citations received, and INPADOC family identifier.

Two features of this extraction are worth making explicit. First, firms are located by the applicant’s address rather than by ownership, so a foreign multinational’s locally-filing subsidiary enters the sample while a domestic firm filing through a foreign holding company does not. The object of interest is innovation carried out by firms located in these economies, which the address criterion captures directly. Second, restricting to corporate applicants excludes individual inventors

and universities, which account for the largest share of granted patents in the region (over 70% in Mexico and Argentina) but are outside the scope of a study of firm responses to trade exposure.

The unit of analysis is the INPADOC patent family, which groups applications of equivalent technical content filed across multiple offices. I define an innovation as an INPADOC family containing at least one granted member, whether granted at the applicant’s home office or abroad.⁵ Defining innovations at the family level is essential to inflated measures: a single invention protected in several jurisdictions would otherwise be counted multiple times, and the degree of multiple filing differs systematically across the three countries. Among granted applications by firms in these countries over 1996–2008, the domestic office accounts for 55.8% in Argentina, 49.6% in Brazil, and 23.7% in Mexico.

A well-known limitation of patent data is that applicant names are entered inconsistently, with varying abbreviations, legal-form suffixes, and misspellings, so that a single firm may appear under several distinct strings and be mistaken for multiple firms. Standardised applicant-name sources such as the OECD’s Harmonised Applicant Names (HAN) database mitigate this, but their coverage of Latin American applicants is limited; I therefore implement a dedicated fuzzy matching procedure. I first apply regular expressions to strip legal-form acronyms (e.g., Ltda., S.A.), city and country names, punctuation, and whitespace, which resolves the most mechanical discrepancies. I then consolidate remaining near-duplicates through manual inspection, using the Jaro–Winkler string distance to flag candidate pairs for review. Manual inspection is difficult to avoid in this setting: legal-form conventions differ across the three countries and many misspellings are idiosyncratic.

Figure A.1 documents the effect of each step. The raw extract contains 3,631 distinct applicant strings associated with 8,937 innovations. Stripping non-distinctive name components resolves 198 of these into existing entities and manual consolidation a further 510, leaving 2,923 firms — a 19.5% reduction in the applicant count. The innovation count is unchanged by construction, since consolidation reassigns applications between firms rather than dropping them. The final step retains firms whose industry can be assigned by the procedure below, giving the estimation sample of 2,576 firms and 8,488 innovations.⁶

⁵The grant requirement raises a question of truncation, since applications filed late in the sample period have had less time to reach grant than early ones. This should not be a concern in the extracted data. Since the window closes in 2008 and the data are drawn from the Autumn 2022 edition, even the latest applications have had fourteen years to be examined.

⁶Appendix A documents the procedure in more detail.

To infer a patenting firm’s industry I follow [Coelli et al. \(2022\)](#) and use the technology classification of its patents. [Aghion et al. \(2016\)](#) show that this patent-based scheme is a legitimate measure, since it reflects the firm’s expectations of where its market will be, and that firms’ patent weights are highly correlated with their sales weights.⁷

For industry assignment I depart from the innovation-counting sample in two ways. First, I use all applications rather than only granted ones: grant status filters for quality in the outcome variable but is uninformative about where a firm directs its inventive effort. Second, I widen the window to 1980–2007, since a firm’s industry is more reliably identified from its full history of applications than from the sample period alone, and pre-sample applications sharpen the assignment for firms patenting sparsely during 1996–2008. As PATSTAT already converts IPC codes into ISIC Rev. 4 industries, I define

$$\theta_{f,i} = \frac{n_{f,i}}{\sum_k n_{f,k}} \quad (1)$$

where $n_{f,i}$ is the number of applications by firm f in industry i over 1980–2007, and assign each firm to the industry carrying its maximum weight.

The resolution at which this assignment can be made varies with the structure of the classification rather than with the quality of a firm’s data. Several manufacturing divisions contain a single 4-digit class, so that the 2- and 4-digit assignments coincide by construction — pharmaceuticals (C21) and tobacco (C12) among them — while others, such as machinery and equipment (C28), subdivide into many. Where a firm’s portfolio identifies a division but does not concentrate within one of its constituent classes, the assignment is made at the 2-digit level and exposure measured for the corresponding aggregate. Of the 2,576 firms in the sample, 456 are assigned at a more aggregate 2-digit level sector.

4.2 Trade data

Measuring exposure to China as a supplier of both competing goods and intermediate inputs requires a bilateral trade dataset at the 4-digit ISIC Rev. 4 level that identifies not only the

⁷Nevertheless, this approach is not entirely neutral. Assigning a firm to an industry through the technology classification of its patents locates the firm in its technology space, which may not coincide with the product-market space in which it competes; [Bloom et al. \(2013\)](#) show that these two dimensions of proximity, while positively correlated, may diverge. The distinction bears most directly on the import-competition channel, which is inherently about product-market rivalry, and least on input access, for which a technology-based assignment is more aligned.

exporting industry but also the importing industry. Cross-country trade databases record the destination country of a flow but not the industry that uses it, so the importing-industry dimension has to be estimated from additional sources.

The starting point is the BACI database (2023 release) developed by CEPII, which harmonises UN COMTRADE records by reconciling CIF-valued imports with FOB-valued exports. BACI is product-level merchandise data; I aggregate it to ISIC Rev. 4 industries using the correspondence tables of the OECD’s Bilateral Trade Database by Industry and End-use (BTDIxE, 2021 edition). These tables serve two purposes: they provide the crosswalk to the ISIC Rev. 4 classification in which firms are assigned an industry, and they split each flow into final and intermediate goods. This yields, for each exporting country x , importing country c , and exporting industry j , a decomposition of total exports into a final demand (consumed by households, governments or invested) and an intermediate component:

$$EXP_{x,c,j} = EXP_{x,c,j}^{FD} + EXP_{x,c,j}^{INP}. \quad (2)$$

The first component supports the import-competition measure. The second requires further decomposition.

I allocate the intermediate component to using industries in two stages. The first uses the OECD’s Inter-Country Input-Output tables (ICIO, 2021 edition), which identify, for a given year, the proportion of country x ’s intermediate exports in industry j absorbed by each 2-digit ISIC importing industry a in country c . ICIO tables cover 68 countries plus a rest of the world aggregate between 1995 and 2018 and are built from national accounts and national input-output or supply-use data. Therefore, these proportions of importing industry vary over time.

The second stage disaggregates the 2-digit industry a into its constituent 4-digit industries i , using national supply-use tables (SUTs): Argentina’s 2004 table, Brazil’s 2011 table, and Mexico’s 2008 table. The share applied is industry i ’s share within a ’s total use of supplying industry j . Only one detailed SUT is available per country, so these shares are time-invariant. Formally:

$$\begin{aligned}
EXP_{x,c,j}^{INP} &= \sum_a EXP_{x,c,j,a} && \text{(ICIO shares, time-varying)} \\
EXP_{x,c,j,a} &= \sum_{i \in a} EXP_{x,c,j,i} && \text{(SUT shares, time-invariant)}
\end{aligned} \tag{3}$$

Assuming stable input structures is common in this literature: [Acemoglu et al. \(2016\)](#) and [Taniguchi \(2019\)](#), studying input access and local labour markets in the United States and Japan respectively, each use a single year of input-output data. The assumption imposed here is narrower than theirs, since it applies only to the composition of 4-digit industries within a 2-digit aggregate, while the allocation across aggregates varies annually through the ICIO. It remains a real limitation, and it is unequal across the three countries: the Argentinian and Mexican tables fall within the sample window, whereas the Brazilian table postdates it by three years and therefore reflects an input structure formed after the shock. No earlier Brazilian supply-use table at this level of detail is available. Still, the shares in question describe which 4-digit activities within a 2-digit division draw on which supplying industries — a feature mainly of structural production technology rather than of sourcing patterns.

Figure 2: From BACI to a country-industry bilateral dataset: an example with Argentinian imports from Chinese machinery (C28) in 2002

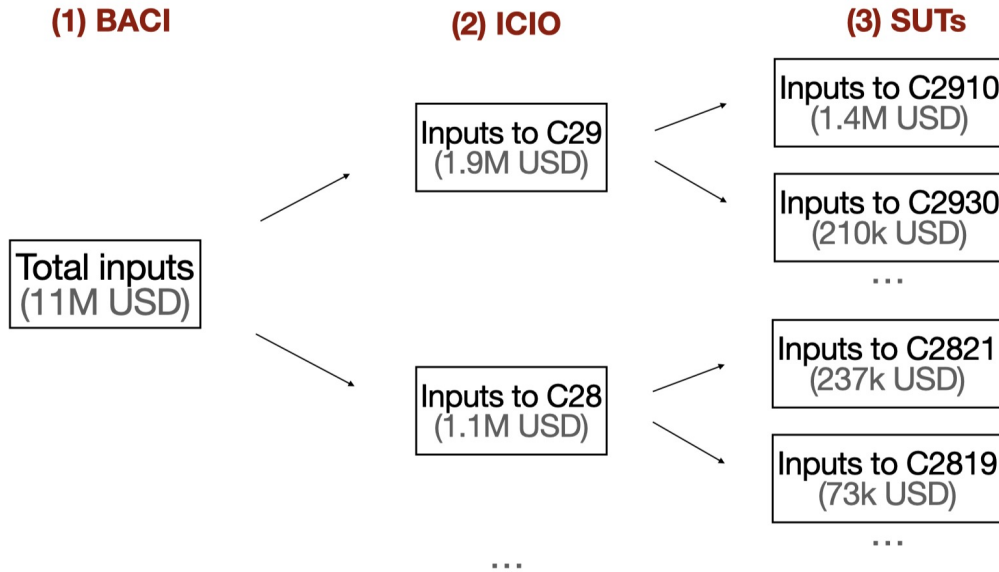


Figure 2 illustrates. BACI and BTDIxE record that in 2002 Argentina imported around USD

11 million worth of intermediates from Chinese machinery (ISIC C28). The ICIO allocates USD 1.9 million of this value to Argentine motor vehicle manufacturing (C29). The Argentine 2004 SUT then attributes USD 210 thousand — about 2% of all Chinese machinery intermediates — to the manufacture of parts and accessories for motor vehicles (C2930).

Two trade flows follow from this construction. Import inflows for industry i is measured by total imports of industry i 's goods, from origin x , irrespective of end use:

$$IMP_{c,i,t}^x = EXP_{x,c,i,t} = EXP_{x,c,i,t}^{FD} + EXP_{x,c,i,t}^{INP}. \quad (4)$$

The intermediate component is retained deliberately. Competition operates in the product market for industry i 's output, and industry i sells to final consumers and to other firms alike: a domestic producer of auto parts competes with Chinese auto parts whether the buyer is a household or a vehicle assembler. Restricting (4) to final demand would define away the competition faced by intermediate-goods producers, who make up a large share of a manufacturing sample.

Input access for industry i is instead measured over the flows into that industry, summed across all supplying industries j , manufacturing and non-manufacturing alike:

$$IMPI_{c,i,t}^x = \sum_j EXP_{x,c,j,i,t}. \quad (5)$$

The two measures are therefore built on the same underlying flows viewed from opposite sides of the transaction: (4) takes industry i as the supplier, (5) as the buyer. A Chinese export surge in a given industry raises competition for the firms producing those goods and improves input access for the firms buying them, and it does so simultaneously. This is not a confound to be removed by construction — doing so would mismeasure both channels — but a reason to estimate the two jointly, so that each coefficient is identified holding the other channel fixed. Section 5 returns to this point.

Constructing the input access instrument requires the same 4-digit resolution for flows outside the treated countries, and here the procedure differs in one respect that should be made explicit. The ICIO provides the 2-digit allocation for the rest-of-the-world aggregate, but no supply-use table exists for such an aggregate, so the second-stage disaggregation cannot be performed on external data. I therefore apply the treated country's own SUT shares to the external flows: the instrument

for country c 's industry i combines rest-of-the-world Chinese intermediate sourcing at the 2-digit level with country c 's own within-division input structure.

4.3 Descriptive statistics

The final dataset comprises 2,576 innovating firms across Argentina, Brazil, and Mexico, responsible for 8,488 innovations between 1996 and 2008, which is the vast majority of patenting activity in Latin America in the period. Brazil alone accounts for 73.7% of firms and 77% of innovations; Argentina and Mexico contribute 13.3% and 13% of firms and 9.4% and 13.6% of innovations respectively.⁸

Table A.2 presents the composition of the sample at the 2-digit industry level, ordered by the change in average yearly patenting between 1996–2001 and 2002–2008. Activity concentrates in four industries — machinery and equipment, other manufactured products, chemicals, and electrical equipment — which together account for 62.8% of firms and 66.7% of innovations; machinery and equipment alone represents just over a quarter of firms and 29.5% of innovations. The ordering by patenting growth does not line up in any simple way with a rise in China's share as a trade partner in Latin American imports and input imports. The largest increases in China's import share occur in leather (39.2 percentage points), textiles (29.6), and ICT and electronics (22.5), industries that sit at opposite ends of the patenting-growth ranking; electrical equipment combines the second-largest import-share increase among the major patenting industries with the second-largest decline in innovation, while chemicals gained most in patenting on comparatively modest exposure. Input exposure is both smaller in magnitude and more evenly distributed, averaging 7.6 percentage points against 12.3 for import competition.

Table A.3 and Figure A.2 report the firm-level distribution. Patenting is extremely concentrated even within this sample of innovators: the median firm produces a single innovation over the thirteen-year window and the 75th percentile two, against a mean of 3.3, and half of all innovations come from 6% of firms. This right-skew is the familiar shape of patent distributions, and it has a direct consequence for the design: 88% of firms have a DHS growth rate – the outcome variable, defined in Section 5 – of exactly -2 or 2 , meaning they either entered or left the pool of innovators between the two periods. Therefore, the variation the empirical strategy exploits is predominantly

⁸Table A.4 reports the country breakdown alongside the corresponding PATSTAT totals for Latin America as a whole.

the extensive margin: firms entering or leaving the pool of innovators, rather than changes in patent counts among firms that patent throughout.

5 Empirical Strategy

The empirical analysis is based on a first-difference specification with country-industry fixed effects (where industry is an industry group), in the spirit of Autor et al. (2020). The unit of observation is a firm; the design compares the change in a firm’s patenting between 1996–2001 and 2002–2008 to the change in its industry’s exposure to China over the same window.

$$\Delta Y_{f,c,i} = \beta_0 + \beta_1 \Delta IMP_{c,i}^{CHN} + \beta_2 \Delta IMPI_{c,i}^{CHN} + \delta_{c,g(i)} + \varepsilon_{f,c,i} \quad (6)$$

Here f indexes firms, $c \in \{\text{Argentina, Brazil, Mexico}\}$ the country in which the firm is located, and i the industry to which the firm is assigned by the procedure of Section 4. Exposure is measured at the finest resolution at which a firm can be placed: for most firms this is a 4-digit ISIC Rev. 4 industry, and for the remainder a 2-digit industry, with the exposure measures aggregated accordingly. The term $\delta_{c,g(i)}$ denotes fixed effects for the interaction of country with an aggregate industry group $g(i)$, following the OECD ICIO’s eight-way grouping of manufacturing.⁹ These absorb any time-invariant country-by-sector properties — such as managerial techniques or a particular high propensity to patent (e.g. the pharmaceutical industry) — while first-differencing removes firm-specific level effects. The coefficients of interest are β_1 and β_2 , the effects of Chinese import competition and Chinese input access respectively.

5.1 Outcome variable

The outcome $\Delta Y_{f,c,i}$ is the Davis–Haltiwanger–Schuh (DHS) growth rate of the firm’s average annual count of innovations:

$$\Delta Y_{f,c,i} = 2 \cdot \frac{\bar{p}_{f,c,i,t} - \bar{p}_{f,c,i,t-1}}{\bar{p}_{f,c,i,t} + \bar{p}_{f,c,i,t-1}} \quad (7)$$

⁹The full mapping is given in Table A.5. These groups are coarser than ISIC’s 34 2-digit manufacturing industries; estimates using 2-digit fixed effects are more volatile, which is unsurprising given the number of Latin American patenting firms per cell.

where $\bar{p}_{f,c,i,t}$ is the firm’s average yearly flow of innovations over 2002–2008 and $\bar{p}_{f,c,i,t-1}$ the corresponding average over 1996–2001. The measure is bounded between -2 and 2 , is symmetric in expansions and contractions, and — crucially in this setting — is defined for firms with zero patenting in either period, taking the value 2 for a firm that begins patenting and -2 for one that stops patenting (Haltiwanger et al., 2013).

This last property is the reason for the choice, and also its principal cost. As Section 4 documents, patenting among Latin American firms is sparse: the median firm in the sample innovates once over the full period, and most of the variation in the data is entry into and exit from the pool of innovators rather than movement in counts among firms already patenting. A log-based outcome would discard exactly this variation, restricting the analysis to a small and highly selected set of continuously-patenting firms. The DHS rate retains it, but does so by assigning a move from zero to one patent a larger value than any increase among incumbents, so the estimates weight the extensive margin heavily. I therefore do not rest the results on this outcome alone. Section 6.3 re-estimates the baseline using the change in patent counts, using patent-weighted regressions that upweight firms with more innovations, and on the subsample of firms patenting in both periods, for which the outcome is purely intensive.

5.2 Measuring exposure

Following Ahn et al. (2018), I measure a country-industry’s exposure to China as the change in China’s weight as a trading partner:

$$\Delta flow_{c,i}^{CHN} = \frac{flow_{c,i,2008}^{CHN}}{flow_{c,i,2008}^{World}} - \frac{flow_{c,i,2002}^{CHN}}{flow_{c,i,2002}^{World}} \quad (8)$$

for $flow \in \{IMP, IMPI\}$: the change in China’s share of industry i ’s imports of competing goods, and of its imported intermediates.

The canonical China-shock measure in this setting typically scales Chinese imports by domestic absorption (Acemoglu et al., 2016; Autor et al., 2020). I treat (8) as a proxy for that object rather than as a substitute for it. Standard Chinese import penetration in industry i can be written as $P_{c,i} = M_{c,i}^{CHN}/D_{c,i}$, where D is domestic absorption, which can be decomposed as $P = \pi \cdot m$, the product of China’s partner share $\pi = M^{CHN}/M$ and the industry’s overall import intensity

$m = M/D$. The change in penetration is then

$$\Delta P = \bar{m} \Delta \pi + \bar{\pi} \Delta m, \quad (9)$$

the sum of a substitution component—China displacing other foreign suppliers within the industry’s existing import bundle—and a scale component, the industry’s overall shift from domestic to foreign sourcing. Equation (8) measures $\Delta \pi$. It therefore captures the first component and omits the second.

Measuring import penetration through the rise of China as a trade partner is known in the literature (see [Bloom et al. \(2016\)](#), for instance), and I adopt it to adapt to data limitations. Constructing $D_{c,i}$ requires domestic production for Argentina, Brazil, and Mexico by 4-digit industry in both 2002 and 2008. Coverage at that resolution is incomplete for all three countries and effectively absent for Argentina, whose 2002 observation would in any case fall in the immediate aftermath of the convertibility crisis described in Section 3. This is why the partner-share approach is the working standard in China-shock research on developing economies: [Iacovone et al. \(2013\)](#) and [Blyde and Fentanes \(2019\)](#) adopt it for this reason, the latter stating explicitly that industry-level production data are unavailable for the Latin American countries in their sample.

5.3 Identification

The exposure measures in (8) are not exogenous to the outcome. A rise in China’s share of Argentinian imports in a given industry may reflect China’s expanding capacity to supply that industry, but it may equally reflect an Argentina-specific shock—a demand expansion, a change in domestic technology, or a policy shift—that drew in imports from the cheapest available source.

Following [Autor et al. \(2020\)](#) and subsequent applications ([Aghion et al., 2024](#); [Ahn et al., 2018](#); [Yamashita and Yamauchi, 2020](#)), I instrument each exposure measure with the corresponding change in China’s partner share in a set of donor economies that excludes the treated country itself:

$$\Delta flow_{O(c),i}^{CHN} = \frac{\sum_{d \in O(c)} flow_{d,i,2008}^{CHN}}{\sum_{d \in O(c)} flow_{d,i,2008}^{World}} - \frac{\sum_{d \in O(c)} flow_{d,i,2002}^{CHN}}{\sum_{d \in O(c)} flow_{d,i,2002}^{World}} \quad (10)$$

where $O(c)$ is the donor set for treated country c and flows are pooled across donors before the share is taken. The logic is the standard one: China’s rise as a supplier to third countries in

industry i reflects the common, China-driven component of the shock—productivity growth, WTO accession, the reallocation of global manufacturing— and is plausibly unrelated to country- c -specific demand or technology shocks affecting the patenting of firms in industry i .

The donor sets are constructed on a leave-one-out basis from a fixed pool of eight Latin American economies: Argentina, Brazil, Chile, Colombia, Mexico, Panama, Peru, and Uruguay. For each treated country c , $O(c)$ comprises the seven remaining economies. Thus Argentina’s import-competition instrument is built from Brazil, Chile, Colombia, Mexico, Panama, Peru, and Uruguay; Brazil’s from Argentina, Chile, Colombia, Mexico, Panama, Peru, and Uruguay; and Mexico’s analogously. The sets differ across the three treated countries only through the leave-one-out exclusion. The pool is regional by design: the identifying assumption is more credible when donors resemble the treated economies in income level, industrial structure, and position relative to the technological frontier, so that China’s penetration of donor markets proxies for the supply shock the treated country faced rather than for an unrelated pattern of demand.

The input-access instrument is constructed differently, since building a partner-share measure for imported intermediates requires input–output structure for the donor economies, and the ICIO and supply-use coverage that Section 4 relies on is not available for the smaller economies in the regional pool. The input instrument therefore draws on a rest-of-the-world aggregate –world flows excluding China and leaving out each treated country—rather than on the regional donor set. The same exclusion logic applies, where a rest-of-the-world aggregate is even less likely than a regional one to share demand shocks with the treated country. The downside is that the donor economies are on average further from the treated countries in industrial structure, so the instrument tracks the treated countries’ input exposure less tightly. This shows up directly in the first-stage diagnostics below, where input access is consistently the weaker of the two channels.

To address potential concerns on the chosen pool of donor countries, in Section 6.3 I re-estimate the baseline with an instrument built from a donor pool of six emerging economies outside Latin America: India, Indonesia, South Africa, Thailand, Turkey and Vietnam. This alternative pool enables a test where the instrument shares no country with the treated sample and no regional demand cycle with it.

In sum, throughout results I estimate (6) by two-stage least squares based on these pools of donor countries. Standard errors are clustered at the ISIC Rev. 4 industry level. Clustering on the industry rather than the country-industry cell follows the structure of the design: the shifters vary

at the industry level, so firms in different countries drawing on the same Chinese industry shock have correlated residuals that country-industry clustering would not capture. This is the level at which shift-share designs are generally understood to require conservative inference (Adão et al., 2019; Borusyak et al., 2022; Goldsmith-Pinkham et al., 2020). Appendix Table A.9 reports the baseline under both schemes.

6 Results

6.1 Baseline estimates

Table 1: Impact of Chinese import penetration and input access on Latin American innovation

	OLS			2SLS		
	(1)	(2)	(3)	(4)	(5)	(6)
ΔIMP	−0.015** (0.007)		−0.016** (0.007)	−0.014* (0.008)		−0.020*** (0.007)
ΔIMPI		0.016 (0.025)	0.025 (0.025)		0.117** (0.050)	0.100*** (0.028)
Num. obs.	2576	2576	2576	2576	2576	2576
Mean outcome variable	0.335	0.335	0.335	0.335	0.335	0.335
Country-sector FE	Yes	Yes	Yes	Yes	Yes	Yes
R ²	0.042	0.038	0.043	0.042	0.018	0.033
KP F				39.3	16.6	13.3

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$. SEs are clustered on 4-digit sectors. Country-industry fixed effects are at an aggregate level, where industries are split into food and tobacco; textiles and apparel; wood, paper and printing; chemicals and non-metallic mineral products; basic and fabricated metals; machinery, equipment and electronics; transport equipment; and other manufacturing. KP F is the Kleibergen–Paap rank Wald F statistic for joint weak identification of the endogenous regressors.

Table 1 reports baseline estimates of equation (6). Columns (1)–(3) present OLS results and columns (4)–(6) their 2SLS counterparts; within each block, both trade shock channels are presented on their own before being estimated jointly. The OLS estimates in columns (1)–(3) associate rising Chinese import competition with lower patenting: a one percentage point (pp.) increase in China’s share of a country-industry’s imports is associated with a decline of roughly 1.5% in average annual patenting.¹⁰ On the other hand, the input access presents a positive but insignificant coefficient. This could reflect the contamination of these estimates by domestic demand: industries whose firms were expanding drew in more imported intermediates for reasons unrelated to the Chinese supply shock, biasing the input-access coefficient toward zero.

¹⁰Bounded between -2 and 2 , the DHS growth rate approaches the conventional percentage change for growth rates near zero. As it approaches its bounds this reading no longer holds: a -50% change corresponds to a -0.67 DHS change, and a $+1$ DHS change translates into a threefold increase.

Correcting for endogeneity concerns, the 2SLS estimates significantly increase the magnitude and significance of the effect of input access on firm innovation. While instrumenting both channels mildly increases the negative effect of a 1pp. Chinese import competition to -2% in average annual patenting, the access to Chinese intermediates emerges as the main driver of patenting activity, consistent with the learning mechanism set out in Section 2: firms operating behind the frontier gain most from intermediates that embody technology they do not themselves produce.

The comparison between the single-channel and joint specifications is itself informative. Moving from column (4) to column (6), the import-competition coefficient grows in absolute value from -0.014 to -0.020 once input access is held constant. The competition effect measured in isolation is therefore attenuated: because the two channels are positively correlated across industries—both originate in the same expansion of Chinese manufacturing exports¹¹—an exposure measure that bundles them nets a discouraging horizontal shock against an enabling vertical one and understates each. This is precisely the confounding that a single import-penetration regressor induces, and it offers a complementary reading of why, beyond what the theory in Aghion et al. (2005) predicts, the literature has not settled on a sign for the innovation response to Chinese imports. The estimated net effect depends on which channel happens to dominate in a given country’s input-output structure.

As reported in Table A.3, the two channels differ substantially in dispersion. Based on coefficients in column (6) and evaluated at one standard deviation of each regressor, import competition implies a change of -0.21 in the DHS growth rate and input access one of $+0.36$. Evaluated instead across the interdecile range actually observed in the sample, the corresponding figures are -0.55 and $+0.73$ ¹². Input access is therefore the larger force on either metric, by a factor of between

¹¹This is particularly sensitive for a sector-level analysis. By nature, sectors, even at the granular ISIC Rev. 4 4-digit level, represent several products. This means that, from the perspective of the focal firm, a greater margin of same-sector imports can reflect complementary inputs rather than competing products. For example, ISIC Rev.4 class 2620 (“Manufacture of computers and peripheral equipment”) includes both computers and components such as storage devices and other computer assemblies. A domestic computer producer may therefore be exposed to 2620 imports within the same 4-digit industry that either compete with its own output (computers) or serve as inputs into its production process (storage devices). Because ΔIMPI aggregates input flows from all supplying industries, including the focal industry itself, such within-industry supply enters both exposure measures: it is captured on the diagonal of the use table as an input to industry i , and simultaneously as an import classified in industry i . The consequence is asymmetric. It leaves ΔIMPI measuring what it is intended to measure, but blends a share of input exposure into ΔIMP , whose coefficient is therefore a weighted average of a negative competition effect and a positive input effect. The competition estimate reported here should be read as a conservative bound: recovering a significant negative coefficient despite this blending implies the underlying horizontal effect is at least as large. Aghion et al. (2024) encounter the same overlap in firm-level customs data—firms frequently export and import within the same detailed product category—and address it by separating their shocks into net export, net import, and common components; product-level data of that kind is not available in this setting, which is why the overlap is bounded rather than removed.

¹²In conventional percentage terms these correspond to changes of -19% and $+44\%$ at one standard deviation,

2.3 and 2.7 in percentage terms, though the margin is considerably narrower than the ratio of coefficients alone would suggest.

Combining the two channels at the exposure levels Latin American manufacturing actually experienced—a 11.5 percentage point rise in China’s share of the region’s manufactured imports, and a 6.8 point rise in its share of imported intermediates—the positive input-access contribution outweighs the negative competition contribution by roughly three to one. The net effect of the China shock on patenting in the region was thus positive, driven by the input side.¹³ That this aggregate ratio exceeds the differential comparisons above is itself informative about how the shock arrived. Input exposure was broad and relatively uniform—a mean of 7.6 points against a standard deviation of 3.6—so most manufacturing firms gained substantial access to Chinese intermediates and few differed sharply from one another. Competition exposure was larger on average but far more unequal, at 12.3 points with a standard deviation of 10.6, concentrating in particular industries. The input channel therefore accounts for more of the region’s aggregate patenting response than of the variation in patenting across firms within it.

6.2 Heterogeneity: which firms and which inputs

The average effects in Table 1 mask considerable heterogeneity. There is widespread evidence of the heterogeneous effects of import competition and market access on firms based on their size, productivity or sales (Aghion et al., 2024; Ahn et al., 2018; Autor et al., 2020). Regardless of the sense of the relationship between these shocks and firm performance, larger, more productive firms are in general better off than their smaller counterparts.

Having established that increased access to imported was a significant driver of firm innovation in Latin America during the China shock, I examine heterogeneous effects along two dimensions. Firstly, through the technological intensity of the focal firm’s sectors, and then through that of its imported inputs.¹⁴ Such an approach enables an overview of where within the economy the response is concentrated and is related to the two main identified mechanisms through which imported intermediates raise innovation: the cost channel and the learning channel.

and -43% and $+115\%$ across the interdecile range.

¹³I emphasise the ordering rather than the implied level: with country-industry fixed effects absorbing the common component of the shock, the coefficients are identified from variation across industries within aggregate groups, and are not intended to support an aggregate counterfactual for regional patenting.

¹⁴This is following a categorization from Eurostat, where manufacturing sectors at the 3-digit level are defined as either high-, medium-high-, medium-low-, and low-technology. I consider a category of high technology, comprising the former two definitions, and low technology, which correspond to the latter two.

Table 2 splits Latin American firms by the technological intensity of the focal firm’s own sector. The 2SLS estimates reveal that the effects of the China shock concentrate around firms in high-technology sectors: firms in these sectors experienced the strongest negative/positive effects on patenting from Chinese import penetration/input access, respectively. At the same time, while the subsample for low-technology firms is naturally smaller in a sample of patenting firms, insignificant results are well identified, with a Kleibergen-Paap statistic of over 35. Additionally, the difference between the two coefficients is itself significant, as reported in the interaction specification in Table A.6.

Table 2: The China shock and industry heterogeneity

	OLS		2SLS	
	(1) Low tech	(2) High tech	(3) Low tech	(4) High tech
ΔIMP	−0.014 (0.009)	−0.016 (0.011)	−0.012 (0.010)	−0.024** (0.011)
$\Delta IMPI$	−0.029 (0.025)	0.059** (0.029)	0.013 (0.012)	0.135*** (0.032)
Num. obs.	779	1797	779	1797
Mean outcome variable	0.228	0.382	0.228	0.382
Country-sector FE	Yes	Yes	Yes	Yes
R ²	0.073	0.038	0.069	0.030
KP F			36.9	11.9

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$. SEs are clustered on 4-digit sectors. Country-industry fixed effects are at an aggregate level, where industries are split into food and tobacco; textiles and apparel; wood, paper and printing; chemicals and non-metallic mineral products; basic and fabricated metals; machinery, equipment and electronics; transport equipment; and other manufacturing. KP F is the Kleibergen–Paap rank Wald F statistic for joint weak identification of the endogenous regressors.

I report the other side of the transaction in Table 3, which separates Chinese input penetration according to the technological intensity of the supplying industry. Each measure records the rise in China’s partner share of a country-industry’s input imports sourced from high- and from low-technology industries respectively, so the two are comparable in units without partitioning the aggregate measure used above.¹⁵ Column (4) reports the effect of high-technology input access alone, and it is both large and the most precisely identified estimate across models: a coefficient of 0.097, significant at the 1% level, with a Kleibergen–Paap F-statistic of 28.

Columns (5) and (6) add low-technology input access and then import competition. The high-

¹⁵Non-manufacturing supplying industries, principally agriculture and extractive activities, are classified as low technology by default, since the Eurostat taxonomy is defined over manufacturing. Input exposure throughout is constructed from merchandise trade and therefore excludes imported services.

Table 3: The China shock and input access heterogeneity

	OLS			2SLS		
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta \text{IMPI}_{\text{HT}}$	0.038*	0.038**	0.047***	0.097***	0.061***	0.069***
	(0.020)	(0.017)	(0.015)	(0.034)	(0.021)	(0.020)
$\Delta \text{IMPI}_{\text{LT}}$		−0.091***	−0.077***		−0.144**	−0.090
		(0.032)	(0.026)		(0.069)	(0.059)
ΔIMP			−0.015***			−0.015***
			(0.004)			(0.005)
Num. obs.	2576	2576	2576	2576	2576	2576
Mean outcome variable	0.335	0.335	0.335	0.335	0.335	0.335
Country-sector FE	Yes	Yes	Yes	Yes	Yes	Yes
R ²	0.040	0.047	0.051	0.033	0.043	0.050
KP F				28.0	7.4	5.4

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$. SEs are clustered on 4-digit sectors. Country-industry fixed effects are at an aggregate level, where industries are split into food and tobacco; textiles and apparel; wood, paper and printing; chemicals and non-metallic mineral products; basic and fabricated metals; machinery, equipment and electronics; transport equipment; and other manufacturing. KP F is the Kleibergen–Paap rank Wald F statistic for joint weak identification of the endogenous regressors.

technology coefficient remains positive and significant throughout, while the low-technology coefficient is negative and loses significance once I also control for import competition. I read the low-technology estimates as suggestive rather than conclusive, since the rest of the world instrument for that flow is much weaker and less relevant, with Kleibergen–Paap statistics of 7.4 and 5.4, so those coefficients are uninformative rather than precisely null.

The OLS estimates in columns (1)–(3) are informative in a different way, since they can be read directly against the baseline. In Table 1, aggregate input access enters positively but insignificantly under OLS, at 0.016 alone and 0.025 jointly. Disaggregating that flow reveals two coefficients of opposite sign and considerable magnitude: high-technology input access at 0.047 and low technology input access at -0.077 , both significant at the 1% level in column (3). The aggregate measure combines these two flows, and its muted OLS coefficient is consistent with the opposite signs towards which they pulling. This highlights the potential endogeneity issues previously discussed, where this influx of low-technology inputs might be correlated to increased demand from less innovative firms. The same logic that motivates separating import competition from input access therefore applies within the input channel itself: an exposure measure that bundles technologically distinct intermediates recovers their net effect and identifies neither. Under instrumentation the aggregate estimate rises to 0.100, close to the 0.097 recovered from high-technology inputs alone in column (4), consistent with the aggregate input-access effect being driven predominantly by the technologically intensive component of the flow.

Taken together, the two tables locate the innovation response with some precision: it runs through technologically sophisticated intermediates and accrues to technologically intensive firms. The two dimensions are largely distinct rather than two readings of the same variation — the correlation between the technological intensity of firm’s sector and its input access exposure to high technology goods is 0.28 — so the results corroborate one another rather than restate a single finding.

This pattern is consistent with the learning interpretation advanced in Section 2, and with the complementarity between input quality and firm capability documented by [Bas and Paunov \(2021\)](#): the gains from technologically advanced intermediates accrue to the firms best placed to absorb what those intermediates embody. It does not, however, decisively exclude a cost interpretation. The China shock expanded the availability of high-technology intermediates to Latin American firms, but also their costs. A firm acquiring same quality, but cheaper Chinese machinery may increase its research effort without learning anything from the technological content of machine itself.

At the same time, the identified pattern does point to a direction where lower costs alone do not explain the increase in Latin American innovation. If this hypothesis were true, one could expect no distinction of the effects based on the technological content of Chinese inputs. The fact that results are driven by high-technology inputs, despite the fact that low-technology inputs represent the larger inflow to Latin America¹⁶, suggests that the learning channel is a significant component in the nexus between Chinese input access and increased Latin American innovation.

6.3 Robustness checks and alternative specifications

This section probes the baseline results along three dimensions, each corresponding to a distinct concern about what the estimates capture. The first concerns the outcome variable: whether the patenting response is an artifact of the DHS transformation and the weight it places on firms entering and leaving the pool of innovators. The second concerns interpretation: whether the increase in patenting reflects genuine inventive activity rather than a change in firms’ propensity to seek legal protection, and what the citation evidence can and cannot establish about the mechanism. The third concerns measurement and identification: whether the results depend on the particular input–output source used to construct input exposure, on the resolution at which firms’ industries

¹⁶Group means computed over the estimation sample; see Table [A.3](#).

are assigned, on the donor pool underlying the instruments, or on the level at which inference is conducted.

Alternative outcome measurements. The baseline outcome is a DHS growth rate, which is defined for firms that did not patent in the pre-period and assigns them the upper bound of +2. Given the mass of firms at zero documented in Section 4, this transformation mechanically places substantial weight on the extensive margin: a move from zero to one patent registers as a larger change than a substantial increase among firms already patenting. A natural concern is that the results reflect entry into and exit from the pool of innovators rather than changes in innovation intensity. I address this with three tests, reported in Table A.7, each of which relaxes a different feature of the baseline outcome.

The first replaces the DHS transformation altogether. Columns (1)–(3) use the change in average yearly patenting between the two periods, an untransformed count difference that imposes no bounds and no symmetric treatment of entry and exit. Input access remains positive and significant at the 1% level, and import competition remains negative, significant at the 10% level in the joint specification. The second test retains the DHS outcome but reduces the influence of firms whose patenting is sparse. Columns (4)–(6) weight observations by the firm’s log patent stock over 1996–2008, following Autor et al. (2020), so that firms with a single patent in thirteen years—which account for a majority of the sample and for most of the mass at the bounds—carry less weight than firms with sustained innovative activity. Both coefficients increase slightly in magnitude relative to the baseline, with input access rising from 0.100 to 0.115. The third test removes the extensive margin by construction. Columns (7)–(9) restrict the sample to the 314 Latin American firms that patented in both 1996–2001 and 2002–2008. For these firms, neither bound of the DHS growth rate is attainable: no observation can take the value +2 or −2, and the mean outcome is close to zero at −0.019. What is therefore captured is the intensive margin of firm innovation. Input access remains positive and significant, at 0.066 alone and 0.043 in the joint specification, and import competition remains negative and significant throughout.

Taken together, these results establish that neither channel is an artifact of the DHS transformation. They also indicate how each channel is distributed across the two margins. Comparing the joint specifications, the input-access coefficient falls from 0.100 in the baseline to 0.043 among always-patenting firms, retaining somewhat under half its magnitude, while the import-competition

coefficient falls from -0.020 to -0.014 , retaining roughly seventy per cent. A meaningful share of the input-access effect therefore operates through new patenting firms, which is itself an economically meaningful response in a region where patenting is scarce and concentrated.

The lawyer effect. Beyond inventive activity, patents may also measure firms' propensity to seek legal protection for existing intellectual property. Because of this, a rise in patenting does not necessarily indicate a rise in innovation. Two versions of this concern apply to the two channels differently, and the estimates in Table A.8 bear on each. Columns (1)–(3) re-estimate the baseline patent specification on the subsample of firms for which citation data are available; columns (4)–(6) report the corresponding estimates with the DHS growth rate of citations per patent as the outcome. Both coefficients on patenting are somewhat larger than in the baseline, indicating this subsample includes more responsive firms to the China shock.

Following Bloom et al. (2016), I use citations per patent as a measure of patent quality and examine its DHS variation, as with the patenting outcomes. The results, reported in Table A.8, add to the evidence that a lawyer effect does not dominate the response of patenting firms to the China shock. On the contrary, I find evidence that increased access to imported intermediates was associated not only with more patenting, but also with higher-quality innovations.

As discussed, there is inherently less reason to expect firms to respond defensively to increased access to intermediates that are complementary, rather than competitive, to their production. I interpret the increase in patent quality beyond the increase in patent quantity through the lens of the literature emphasising the role of imported intermediates in expanding firms' technological opportunities. In particular, access to new imported inputs can expand the set of technologies and products available to domestic firms, enabling innovations that were previously infeasible (Goldberg et al., 2010). Consistent with the learning channel, imported intermediates may therefore act as complements to firms' internal innovative capabilities, rather than merely reducing production costs. This interpretation is also consistent with the broader literature on trade-induced quality upgrading, which emphasises how access to foreign markets and inputs can enable firms to undertake the complementary investments required to produce higher-quality goods (Verhoogen, 2008). The resulting innovations may therefore be not only more numerous, but also more technologically ambitious and consequential. The increase in patent citations is consistent with this interpretation, suggesting that the China shock may have improved the quality and technological impact of firms' innovations, rather than merely inducing additional patenting.

Alternative specifications. Baseline estimates rest on a particular construction of exposure, a particular donor pool for the instruments, and a particular level of inference. A last group of robustness checks therefore asks whether results depend on those choices.

Firstly, I examine the relationship on which the instrumental-variables estimates are built. Table A.10 reports reduced-form regressions of the outcome directly on the two instruments, alongside the corresponding 2SLS estimates. Reduced-form inference does not depend on first-stage strength, so these estimates provide a check on the main results that is unaffected by the identification concerns discussed in Section 5. The instrument for input access enters positively and significantly at the 1% level, both alone and jointly, confirming that the input-access result reflects a genuine relationship between the China-driven component of input exposure and firm patenting rather than an artefact of instrumentation. The instrument for import competition is weaker, entering negatively but significantly only at the 10% level in the joint specification. This asymmetry recurs throughout the paper: the input-access result is the more robust of the two, and the competition result, while consistently signed, is estimated with less precision.

A second concern relates to inference rather than estimation. As Adão et al. (2019) and Borusyak et al. (2022) show, standard errors in shift-share designs can understate uncertainty when identifying variation comes from a limited number of common shocks, since residuals are correlated across units with similar exposure. The baseline clusters standard errors at the level at which the shocks vary. I have re-estimated the baseline clustering instead at the 2-digit industry level and at the country-industry (4-digit) level; no coefficient changes its significance classification under either scheme.

Turning to measurement, the input-access measure rests on a specific input–output source. Table A.11 re-estimates the input-access effect using the World Input–Output Database in place of the OECD’s ICIO, applying the same national supply-use tables and the same disaggregation procedure to reach the 4-digit level. WIOD is designed to track developments over time and is benchmarked to national accounts time series (Timmer et al., 2015), making it a natural alternative. Because WIOD does not cover Argentina, the comparison is conducted on the 2,233 Brazilian and Mexican firms for which both measures are available. The disaggregation to ISIC 4-digit bilateral trade estimations is performed using the same supply and use tables, following the same procedure as the baseline model, so discrepancies reflect the difference between the estimation approaches to reach the multi-regional input-output tables. While the individual coefficient for input access

is weaker in magnitude in WIOD vis-à-vis the ICIO, the workhorse model is remarkably similar across both measurements, reinforcing the baseline results.

Table A.12 varies the donor pool underlying the instruments. In place of other Latin American economies for import competition and a rest-of-world aggregate for input access, I construct both instruments from a basket of six emerging economies outside the region: India, Indonesia, South Africa, Thailand, Turkey, and Vietnam. The rationale is to isolate the China-driven component of exposure from any regional shock common to Latin America, which the baseline donor pool by construction cannot rule out. The estimates are consistent with the baseline in sign and broadly in magnitude, with import competition at -0.020 and input access at 0.125 in the joint specification. However, the alternative instruments are considerably weaker: the Kleibergen–Paap statistic in the joint specification falls to 5.8 , well below conventional thresholds. It indicates that the results do not depend on the regional character of the donor pool, but the alternative instruments are not strong enough to support inference in their own right, which is why the baseline estimates is preferred.

7 Conclusion

This paper aims to analyse how the China shock affected firm innovation in Latin America and argues that the question cannot be answered without separating the two ways in which a single expansion of Chinese exports reaches a firm. Chinese goods arrive in the product market a firm sells into as and they arrive in the input market it buys from, and the two carry opposite predictions. Using bilateral trade data layered with inter-country and national input–output tables, I construct both exposures at the 4-digit industry level for Argentina, Brazil, and Mexico, and estimate their effects jointly on the patenting of 2,576 innovating firms between 1996 and 2008.

The two channels pull in opposite directions, and the vertical one dominates. Instrumenting each exposure with China’s rise in donor economies, a one percentage point increase in China’s share of a country-industry’s imports reduces average annual patenting by around 2%, while an equivalent increase in China’s share of its imported intermediates raises it by around 10%. The ratio of coefficients overstates the difference in economic terms, since the two exposures are distributed very differently across industries: evaluated at one standard deviation of each regressor, input access is the larger force by a factor of roughly two and a half. Evaluated instead at the exposure

levels Latin American manufacturing actually experienced over 2002–2008 from the China shock, the input-side contribution outweighs the competition-side contribution by three to one. Across a series of robustness checks and alternative measures, this finding remains consistent: the net effect of the China shock on patenting in the region ran through the input side and was positive.

Both channels concentrate among technologically intensive firms, and the input channel runs through technologically intensive inputs. Splitting firms by the technological intensity of their own sector, the estimates for high-technology firms are larger in absolute value on both channels, and the difference between the two groups is itself significant. Splitting input exposure by the technological intensity of the supplying industry, the effect is carried entirely by high technology intermediates. These are largely independent tests rather than two readings of the same variation, since a firm’s own technological intensity and the relative intensity of its input exposure are only weakly correlated in the sample. The pattern is consistent with the learning interpretation: firms operating well behind the global frontier gain most from intermediates that embody technology they do not themselves produce, and the gains accrue to the firms best placed to absorb what those intermediates carry. It does not decisively exclude a cost interpretation, since cheaper high-technology inputs and learning from high-technology inputs are observationally similar here. What the evidence seems to rule out is a pure cost story, which would predict no distinction by the technological content of the inputs: low-technology intermediates were the larger inflow to the region and carry no measurable patenting response. The citation results point in the same direction, indicating that the additional patenting was not simply defensive legal protection of existing intellectual property.

Two contributions follow. The first is evidence on input access in an emerging economy setting, where the channel has been studied far less than import competition and where theory predicts it should matter most. [Aghion et al. \(2024\)](#) recover a positive but not consistently significant input-supply effect for French firms; in Latin America the corresponding effect is larger, robust, and the dominant force in the data. The distance-to-frontier logic that makes competition discouraging in this setting is the same logic that makes input access enabling, and the region therefore separates the two channels more sharply than the developed economies in which they have mostly been studied. The second contribution is a reading of why the literature has not settled on a sign for the innovation response to Chinese imports. Beyond the competing Schumpeterian and escape-competition forces, a single import-penetration regressor bundles a discouraging horizontal shock with an enabling vertical one, and recovers their net effect while identifying neither. The estimates

here show this directly: the competition coefficient grows in absolute value once input access is held constant. Which channel dominates a bundled measure depends on a country’s input–output structure and on its position in the China trade relationship, and differences of that kind can account for a substantial part of the contrast between existing estimates.

These results must be contextualised within their limitations. Exposure is measured as the change in China’s weight as a trading partner rather than as a change in import penetration scaled by domestic absorption, a choice imposed by the absence of 4-digit production data for these economies and shared with other work on developing countries; it captures the substitution of Chinese for other foreign suppliers and omits the shift from domestic to foreign sourcing. Patenting is an imperfect proxy for innovation in a region where it is scarce and concentrated, and the design’s identifying variation is predominantly the entry and exit of firms from the pool of innovators, though the results survive outcome definitions that remove this feature. And the input-access instrument, drawn from a rest-of-the-world aggregate rather than a regional donor pool, tracks treated-country exposure less tightly than its competition counterpart, which is visible in the first-stage diagnostics throughout.

The China shock episode in Latin America itself is best understood as a commodities-for-manufactures trade boom. China’s demand shock to the region fell on commodity producers rather than on the manufacturing firms that patent, which is why the export margin bears so little on the question and why the paper sets it aside. What reached manufacturing was an import surge that arrived as competition and as supply at once, and the two are not separable in the aggregate penetration measures through which the shock is usually read.

That the supply side dominated carries a policy reading and a limit. The reading is that restricting access to imported intermediates in order to shield domestic producers works against the channel through which this shock raised innovation, and does so most sharply for the technologically intensive firms that account for the response. The limit is that those firms were already equipped to absorb what foreign inputs embody, consistent with the broader finding that complementary capabilities, rather than access to frontier technology itself, are what bind in developing economies (Cirera and Maloney, 2017). Better inputs likely raised innovation among firms able to use them, rather than spurring the creation of such firms.

Manufacturing in developed and emerging economies is conventionally cast on the losing side of the China shock. However, for firms in emerging economies that have remained significantly

behind the productivity frontier, the inflow of Chinese goods – already in the early 2000s – was something to learn and lower costs from as much as something to compete with. Among Latin America’s innovating firms, I find that it is the former that dominates.

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A Data Appendix: Patent data construction

This appendix documents the construction of the firm-level patent sample, from the raw PATSTAT extract to the estimation sample, and discusses the nature of the firms lost at each stage. Figure A.1 summarizes the process. The key point is that the reduction in firms comes from two conceptually distinct sources: the consolidation of duplicate firm identities, which removes spurious firms without discarding any innovations, and the imposition of data requirements on sector information, which removes genuine firms. Distinguishing the two is important for interpreting the final sample.

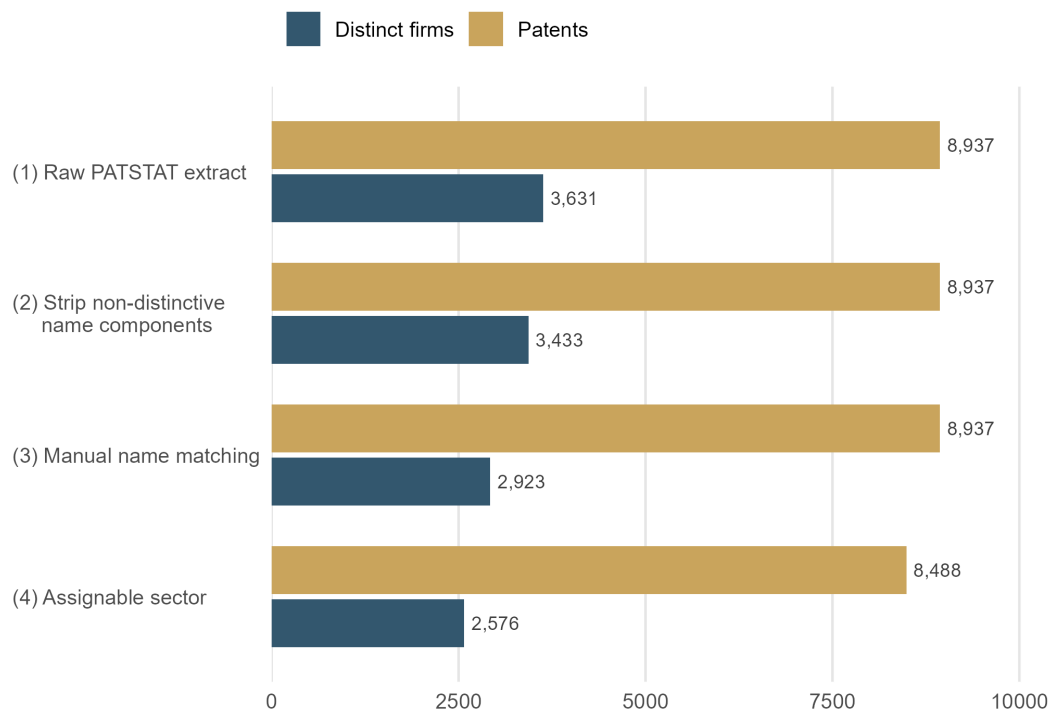
Deduplication (stages 1–3). The raw extract comprises 3,631 distinct applicant strings located in Argentina, Brazil, or Mexico, associated with 8,937 INPADOC patent families. Stages 2 and 3 address the well-known problem that a single firm appears under many name variants. Stage 2 strips non-distinctive components through regular expressions¹⁷, and stage 3 consolidates the remaining near-duplicates through manual inspection supported by the Jaro–Winkler string distance. Together, these stages reduce the count from 3,631 to 2,923 firms while leaving the number of patents unchanged at 8,937: no innovation is discarded, and the apparent loss of firms reflects the recognition that several strings corresponded to the same entity. Consistent with this, the average number of patents per firm rises from 2.5 to 3.1 over these stages, as patents are reassigned from spurious duplicates to their true holders.

The scale of the consolidation reflects how heavily a single firm can be fragmented across raw strings. Table A.1 illustrates this with the case of Petrobras, which appears in the raw data under [N] distinct strings owing to abbreviations, transcription errors, and inconsistent rendering of its full legal name. The example also conveys why a purely automated procedure is insufficient: several variants are not recoverable through string distance alone and require manual judgment, while conversely some genuinely distinct firms have deceptively similar names.

¹⁷The removed components fall into three groups, none of which conveys firm identity. The first is legal-form and corporate descriptors, in their Spanish, Portuguese, and English variants and common abbreviations: for example *Sociedad Anónima*, *S.A.*, *S/A*, *Limitada*, *Ltda.*, *Compañía*, *Companhia*, *Cia.*, *Corporation*, *Inc.*, *Holding*, and the various extended Argentine forms (e.g. *S.A.I.C.F.*, *S.A.I.F.I.A.*) and Mexican *de Capital Variable (de C.V.)*. The second is geographic terms—country and major-city names such as *Brasil*, *México*, *Perú*, *Buenos Aires*, *São Paulo*, and *Ciudad de México*—which frequently appear in applicant names but do not distinguish firms. The third is generic line-of-business descriptors, again across languages, such as *Industria(s)*, *Industrial(es)*, *Comercial*, *Comércio*, *Importação*, *Exportação*, and *Financiera*, together with conjunctions (*e*, *y*, and). Each term is removed only as a standalone word—that is, when bounded by spaces—so that substrings inside genuine firm names are preserved.

Sector requirements (stages 4–5). The final stage imposes the data requirements needed for the empirical analysis. Stage 4 retains firms for which a sector was assigned. The reduction of the sample to 2,576 firms and 8,488 patents reflects the removal of non-manufacturing firms from the sample. Most of the removed firms operated in the construction and computer programming sectors.

Figure A.1: Construction of the patent sample



Note: The figure reports the number of distinct firms and INPADOC patent families retained at each stage of the sample construction. Stages (1)–(3) are deduplication steps that consolidate name variants without discarding patents; stages (4)–(5) impose sector-information requirements. (1) Raw PATSTAT extract of company applicants in Argentina, Brazil, and Mexico; (2) removal of non-distinctive name components; (3) manual name matching; (4) firms with an assignable sector.

Table A.1: Illustration of name variation: the case of Petrobras

<i>Source of variation</i>	<i>Raw applicant string (examples)</i>
Canonical name	PETROLEO BRASILEIRO S.A. - PETROBRAS
Punctuation and legal form	PETROLEO BRASILEIRO S/A - PETROBRAS PETROLEO BRASILEIRO S.A. PETROLEO BRASILEIRO PETROBRAS
Spelling and transcription errors	PETROLEW BRASILEIRO S.A. - PETROBRAS PETROLEO BASILEIRO S.A. - PETROBRAS PETROLEO BRISILEIRO S.A. - PETROBRAS PETROLEO BRASILEIRO SA - PETROBRSS
Character-encoding / OCR artifacts	PETR[.]LEO BRASILEIRO SA - PETROBR[.]S PETROLEO BRASILEIRO S A - PETROBR[.]S
Non-Portuguese renderings	PETROLEUM BRAZIL BRAZIL PETROLEUM COMPANY PETOROREO BURAJIREIRO SA PETOROBURASU
Appended tokens (location, notes)	... - PETROBRAS, RIO DE JANEIRO ... - PETROBRAS (FIRMA)

Note: Examples of distinct raw applicant strings in PATSTAT that all correspond to the single firm Petrobras. Punctuation and legal-form variants are resolved automatically; spelling errors are flagged by string-distance comparison; encoding artifacts and non-Portuguese renderings (including English translations and a Japanese transliteration) carry low string similarity to the canonical name and are recoverable only through manual inspection.

B Tables Appendix

Table A.2: Final sample by industry

Industry (ISIC 2-d)	(1) Firms _n	(2) Firms _%	(3) Pat _n	(4) Pat _%	(5) ΔPat_{yavg}	(6) ΔIMP	(7) ΔIMPI
Chemicals	240	9.3%	840	9.9%	47.3	4.0	2.3
Pharmaceuticals	166	6.4%	492	5.8%	29.4	2.1	6.6
ICT and electronics	181	7%	376	4.4%	24.3	22.5	4.9
Machinery and equipment	681	26.4%	2508	29.5%	20.0	9.8	7.2
Oth. transport equipment	47	1.8%	176	2.1%	11.2	5.5	9.3
Food prod.	68	2.6%	144	1.7%	10.7	2.4	2.4
Fabricated metal prod.	79	3.1%	225	2.7%	7.7	14.6	7.3
Oth. manufactured prod.	481	18.7%	1245	14.7%	7.6	14.0	8.7
Furniture	54	2.1%	149	1.8%	3.6	20.2	8.0
Leather and related	18	0.7%	77	0.9%	3.3	39.2	7.7
Motor vehicles	135	5.2%	413	4.9%	2.4	2.7	12.8
Printing and reproduction	7	0.3%	9	0.1%	0.7	12.8	5.5
Textiles	8	0.3%	8	0.1%	0.5	29.6	5.5
Beverages	4	0.2%	6	0.1%	0.2	5.8	9.2
Paper prod.	12	0.5%	27	0.3%	0.1	2.8	6.0
Oth. non-metal mineral prod.	60	2.3%	170	2%	0.1	15.9	3.3
Coke and petroleum	4	0.2%	4	0%	0.0	-0.6	0.5
Wearing apparel	7	0.3%	10	0.1%	-0.1	9.3	22.3
Wood prod.	3	0.1%	3	0%	-0.2	10.3	9.5
Rubber and plastics	50	1.9%	96	1.1%	-1.1	10.0	8.1
Tobacco prod.	1	0%	22	0.3%	-1.2	0.2	8.6
Electrical equipment	215	8.3%	1065	12.5%	-3.5	21.4	9.8
Basic metals	55	2.1%	423	5%	-14.5	9.5	-3.7
Total manufacturing	2576	100%	8488	100%	148.4	11.5	6.8

Note: The table is in descending order of the variation in average yearly patents (column 5). This variation corresponds to the difference between the average yearly patents in 1996-2001 and 2002-2008. The last two deltas correspond to the change between 2002 and 2008 in China's partner share of Argentina's, Brazil's and Mexico's combined imports and input imports. Industries (ISIC 2-digit) that did not report any patenting between 1996 and 2008 are not reported.

Table A.3: Summary of sample distribution at the firm level

Variable	(1) Mean	(2) SD	(3) 1st	(4) 10th	(5) 25th	(6) 50th	(7) 75th	(8) 90th	(9) 99th	(10) 99.9th
Pat	3.3	14.4	1.0	1.0	1.0	1.0	2.0	5.0	34.5	144.0
Δ Pat	0.3	1.9	-2.0	-2.0	-2.0	2.0	2.0	2.0	2.0	2.0
Cit	25.0	176.7	0.0	0.0	0.0	0.0	10.0	38.0	433.8	2620.8
Δ Cit	0.2	1.2	-2.0	-2.0	0.0	0.0	1.3	2.0	2.0	2.0
Δ IMP	12.3	10.6	-0.3	2.3	3.9	8.9	16.3	29.6	40.3	43.7
Δ IMPI	7.6	3.6	-4.3	3.5	5.5	7.8	9.2	10.8	15.9	30.4

Note: Deltas for outcome variables patents (column 2) and citations (column 4) follow the specification of the DHS growth rate used in the empirical strategy. The DHS growth rate is symmetric and allows for defining variation of entry (i.e., started patenting, defined at 2) and exit (i.e., stopped patenting, defined at -2).

Table A.4: Distribution of patenting firms and patents by Latin American country, 1996-2008

Country	PATSTAT				Final sample			
	Firms _n	Firms%	Pat _n	Pat%	Firms _n	Firms%	Pat _n	Pat%
Argentina	623	14.5%	1047	10.2%	343	13.3%	794	9.4%
Brazil	2549	59.3%	6686	65.3%	1898	73.7%	6537	77%
Mexico	459	10.7%	1204	11.8%	335	13%	1157	13.6%
LATAM3	3631	84.6%	8937	86.9%	2576	100%	8488	100%
Rest of LATAM	660	15.4%	1346	13.1%	-	-	-	-
LATAM	4291	100%	10283	100%	-	-	-	-

Source: PATSTAT

Notes: The difference between PATSTAT and final sample in values for Argentina, Brazil and Mexico arise from multiple steps in data processing. Firstly, the number of firms from PATSTAT is overestimated due to different names and typos in the dataset, which are treated through name matching exercises described in the data section. Secondly, the industry cannot be accurately estimated for all firms, due to eventually incomplete data in PATSTAT. Finally, a small portion of these firms are innovating outside manufacturing and are not part of the object of this study.

Table A.5: The eight industry aggregates under manufacturing

Industry aggregate	ISIC 2-digit industries
Food and tobacco	10 to 12
Textiles and apparel	13 to 15
Wood, paper and printing	16 to 18
Chemicals and non-metallic mineral products	19 to 23
Basic and fabricated metals	24 to 25
Machinery, equipment and electronics	26 to 28
Transport equipment	29 to 30
Other manufacturing	31 to 33

Note: Industries aggregates derived from the breakdown provided by the ICIO-derived (Inter Country Input-Output Tables) Trade in Value Added database (oe.cd/tiva).

Figure A.2: Distribution of firm total patent stock and DHS delta in the sample

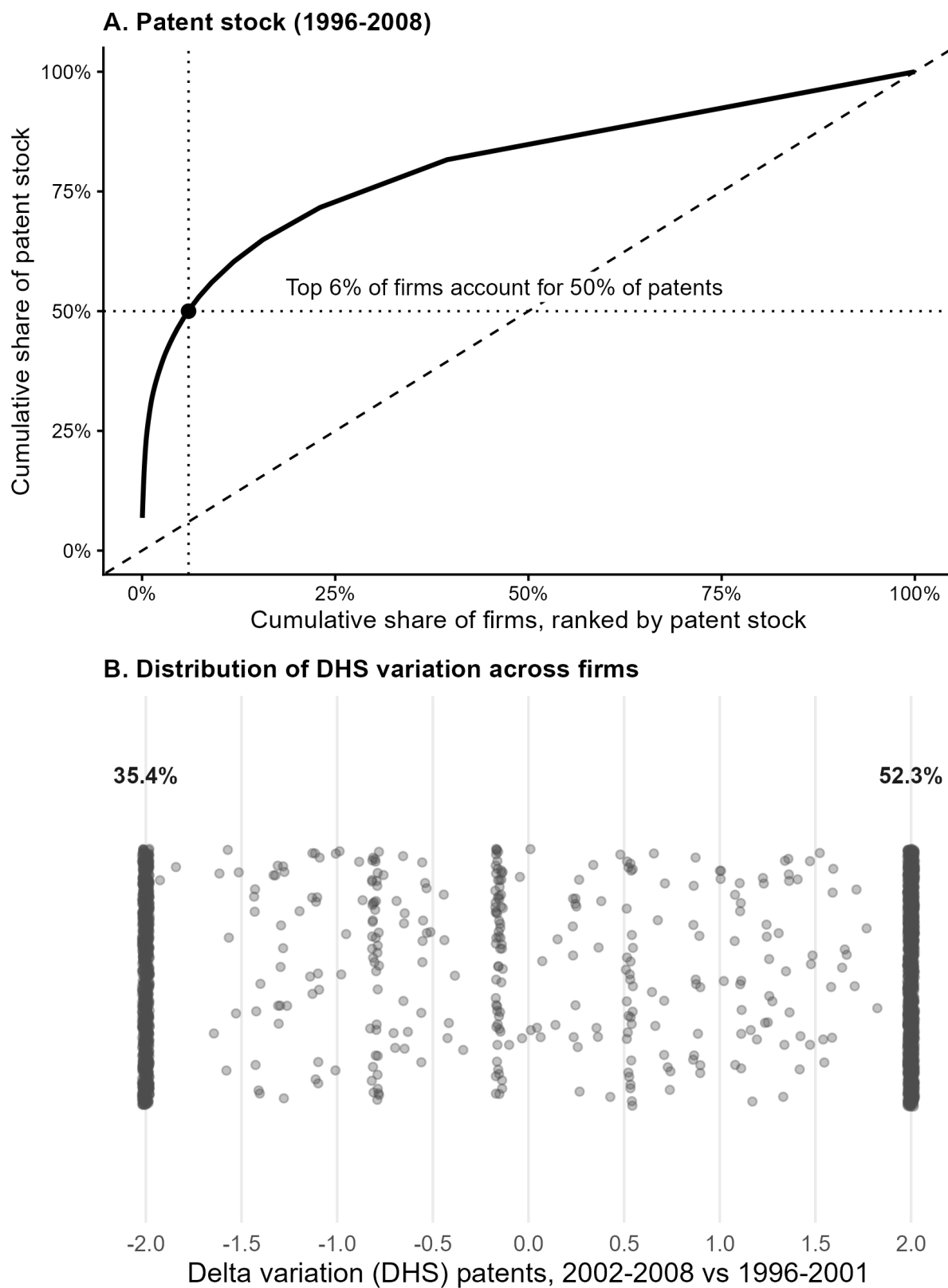


Table A.6: The China shock and industry heterogeneity: interaction specification

	(1) 2SLS
ΔIMP	−0.001 (0.013)
ΔIMPI	0.051 (0.032)
$\Delta\text{IMP} \times \text{High tech}$	−0.023 (0.018)
$\Delta\text{IMPI} \times \text{High tech}$	0.070* (0.038)
Num. obs.	2576
Mean outcome variable	0.335
Country-sector FE	Yes
R ²	0.050
KP F-stat. (first-stage)	21.3

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$. SEs are clustered on 4-digit sectors. Country-industry fixed effects are at an aggregate level, where industries are split into food and tobacco; textiles and apparel; wood, paper and printing; chemicals and non-metallic mineral products; basic and fabricated metals; machinery, equipment and electronics; transport equipment; and other manufacturing.

Table A.7: Robustness: patent counts, patent-weighting, and the always-patenting subsample

	Patent counts (2SLS)			Patent-weighted (2SLS)			Always-patenting subsample (2SLS)		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
ΔIMP	−0.002 (0.002)		−0.004* (0.002)	−0.014* (0.008)		−0.022*** (0.007)	−0.011* (0.006)		−0.014** (0.006)
ΔIMPI		0.036*** (0.010)	0.033*** (0.007)		0.138** (0.054)	0.115*** (0.025)		0.066** (0.030)	0.043* (0.024)
Num. obs.	2576	2576	2576	2576	2576	2576	314	314	314
Mean outcome variable	0.058	0.058	0.058	0.347	0.347	0.347	−0.019	−0.019	−0.019
Country-sector FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R ²	0.009	0.012	0.012	0.044	0.015	0.035	0.108	0.109	0.114
KP F	39.3	16.6	13.3	57.4	13.3	18.5	29.5	11.5	16.6

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$. SEs are clustered on 4-digit sectors. Country-industry fixed effects are at an aggregate level, where industries are split into food and tobacco; textiles and apparel; wood, paper and printing; chemicals and non-metallic mineral products; basic and fabricated metals; machinery, equipment and electronics; transport equipment; and other manufacturing. Columns (1)–(3) use the change in average yearly patenting between the two periods as the outcome variable; columns (4)–(9) use the DHS growth rate, as in the baseline specification. Columns (4)–(6) are weighted by the log stock of patents of a firm between 1996 and 2008. The always-patenting subsample in columns (7)–(9) includes only firms with at least one patent in both the 1996–2001 and 2002–2008 periods.

Table A.8: Robustness: patents and citations per patent on the citations-available subsample

	Patents			Citations per patent		
	(1)	(2)	(3)	(4)	(5)	(6)
ΔIMP	-0.014*		-0.023***	-0.000		-0.004
	(0.008)		(0.008)	(0.004)		(0.003)
ΔIMPI		0.165*	0.131***		0.054**	0.048**
		(0.083)	(0.032)		(0.026)	(0.020)
Num. obs.	1842	1842	1842	1842	1842	1842
Mean outcome variable	0.260	0.260	0.260	0.230	0.230	0.230
Country-sector FE	Yes	Yes	Yes	Yes	Yes	Yes
R ²	0.038	0.047	0.049	0.009	0.013	0.013
KP F	34.9	10.9	12.9	34.9	10.9	12.9

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$. SEs are clustered on 4-digit sectors. Country-industry fixed effects are at an aggregate level, where industries are split into food and tobacco; textiles and apparel; wood, paper and printing; chemicals and non-metallic mineral products; basic and fabricated metals; machinery, equipment and electronics; transport equipment; and other manufacturing. Columns (1)–(3) re-estimate the baseline patent specification on the citations-available subsample, since this differs from the main estimation sample; columns (4)–(6) report the corresponding citations-per-patent estimates on the same subsample.

Table A.9: Robustness: alternative clustering schemes

	OLS			2SLS		
	(1)	(2)	(3)	(4)	(5)	(6)
ΔIMP	-0.014*		-0.020***	-0.014*		-0.020***
	(0.007)		(0.006)	(0.007)		(0.007)
ΔIMPI		0.117**	0.100***		0.117**	0.100***
		(0.049)	(0.025)		(0.047)	(0.029)
Num. obs.	2576	2576	2576	2576	2576	2576
Mean outcome variable	0.335	0.335	0.335	0.335	0.335	0.335
Country-sector FE	Yes	Yes	Yes	Yes	Yes	Yes
SE cluster	ISIC 2-digit	ISIC 2-digit	ISIC 2-digit	Cty-ISIC 4-digit	Cty-ISIC 4-digit	Cty-ISIC 4-digit
R ²	0.040	0.047	0.048	0.040	0.047	0.048
KP F	38.9	16.4	12.9	38.9	16.4	12.9

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$. Country-industry fixed effects are at an aggregate level, where industries are split into food and tobacco; textiles and apparel; wood, paper and printing; chemicals and non-metallic mineral products; basic and fabricated metals; machinery, equipment and electronics; transport equipment; and other manufacturing.

Table A.10: Robustness: reduced-form estimates

	2SLS			Reduced-form OLS		
	(1)	(2)	(3)	(4)	(5)	(6)
ΔIMP	-0.007 (0.012)		-0.020** (0.008)			
$\Delta IMPI$		0.117** (0.050)	0.100*** (0.030)			
Z_{IMP}				-0.018 (0.011)		-0.013* (0.007)
Z_{IMPI}					0.158*** (0.042)	0.147*** (0.036)
Num. obs.	2576	2576	2576	2576	2576	2576
Mean outcome variable	0.335	0.335	0.335	0.335	0.335	0.335
Country-sector FE	Yes	Yes	Yes	Yes	Yes	Yes
R ²	0.038	0.047	0.049	0.040	0.047	0.048
KP F	38.9	16.4	12.9			

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$. SEs are clustered on 4-digit sectors. Country-industry fixed effects are at an aggregate level, where industries are split into food and tobacco; textiles and apparel; wood, paper and printing; chemicals and non-metallic mineral products; basic and fabricated metals; machinery, equipment and electronics; transport equipment; and other manufacturing. Z_{IMP} is the instrument based on the Latin American countries basket; Z_{IMPI} is the instrument based on the rest-of-world (ROW) basket. Columns (4)–(6) report reduced-form OLS estimates using these instruments directly as regressors.

Table A.11: The input access effect, calibrated with the ICIO and WIOD

	ICIO		WIOD	
	(1)	(2)	(3)	(4)
$\Delta IMPI$	0.101** (0.050)	0.084*** (0.029)	0.069* (0.036)	0.076** (0.035)
ΔIMP		-0.018** (0.007)		-0.017** (0.007)
Num. obs.	2233	2233	2233	2233
Mean outcome variable	0.428	0.428	0.428	0.428
Country-sector FE	Yes	Yes	Yes	Yes
R ²	0.003	0.017	0.013	0.019
KP F	14.6	17.6	54.0	24.9

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$. SEs are clustered on 4-digit sectors. Country-industry fixed effects are at an aggregate level, where industries are split into food and tobacco; textiles and apparel; wood, paper and printing; chemicals and non-metallic mineral products; basic and fabricated metals; machinery, equipment and electronics; transport equipment; and other manufacturing. KP F is the Kleibergen–Paap rank Wald F statistic for joint weak identification of the endogenous regressors.

Table A.12: Alternative basket of countries: six non-Latin American emerging economies

	Baseline basket			Alternative basket		
	(1)	(2)	(3)	(4)	(5)	(6)
ΔIMP	-0.014^* (0.008)		-0.020^{***} (0.007)	-0.017^* (0.010)		-0.020^{***} (0.007)
$\Delta IMPI$		0.117^{**} (0.050)	0.100^{***} (0.028)		0.080^* (0.042)	0.125^{***} (0.042)
Num. obs.	2576	2576	2576	2576	2576	2576
Mean outcome variable	0.335	0.335	0.335	0.335	0.335	0.335
Country-sector FE	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.042	0.018	0.033	0.035	0.030	0.034
KP F	39.3	16.6	13.3	33.9	11.2	5.8

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$. SEs are clustered on 4-digit sectors. Country-industry fixed effects are at an aggregate level, where industries are split into food and tobacco; textiles and apparel; wood, paper and printing; chemicals and non-metallic mineral products; basic and fabricated metals; machinery, equipment and electronics; transport equipment; and other manufacturing. The alternative basket of countries for the instrumental variable is composed of India, Indonesia, South Africa, Thailand, Turkey and Vietnam. KP F is the Kleibergen–Paap rank Wald F statistic for joint weak identification of the endogenous regressors.